

Deployment



Crear el repositorio

1. Crear el repositorio
2. Vamos al apartado de [Actions](#), donde vamos a encontrar un “marketplace” con distintos sets de configuraciones para cada necesidad, como un action ya construido para Node.js, sin necesidad de crear algo ad hoc.
3. Seleccionamos la opción de Node.js, clic en Configure
4. Nos va a redirigir a una página de edición de un archivo .yml donde está cargada toda la configuración básica para correr un pipeline de Node.js, el mismo incluye:
 - a. Descargar dependencias
 - b. Buildear la aplicación
 - c. Correr los tests
5. Al comienzo no es necesario modificar nada, hacemos clic en “Commit changes...” y se nos va a crear en nuestra rama principal, una carpeta .github/workflows con un archivo node.js.yml

Choose a workflow

Build, test, and deploy your code. Make code reviews, branch management, and issue triaging work the way you want. Select a workflow to get started.

Skip this and [set up a workflow yourself](#) →

Categories

Deployment

Security

Continuous integration

Automation

Pages

🔍 node

Found 13 workflows

Deploy Node.js to Azure Web App



By Microsoft Azure

Build a Node.js project and deploy it to an Azure Web App.

Configure

Deployment

Deploy Node.js to Azure Functions App



By Microsoft Azure

Build a Node.js project and deploy it to an Azure Functions App on Windows or Linux.

Configure

Deployment

Grunt



By GitHub Actions

Build a NodeJS project with npm and grunt.

Configure

JavaScript

Gulp



By GitHub Actions

Build a NodeJS project with npm and gulp.

Configure

JavaScript

Webpack



By GitHub Actions

Build a NodeJS project with npm and webpack.

Configure

JavaScript

njsscan



By NodeJSScan

nodejsscan is a static security code scanner that finds insecure code patterns in your Node.js applications.

Configure

Code scanning

Marketplace

ConfigureJavaScript

NowSecure Mobile SBOM
By NowSecure
Generate a Mobile SBOM for an application and submit to Dependency Graph
ConfigureCode scanning

ConfigureJavaScript

NowSecure
By NowSecure
The NowSecure Action delivers fast, accurate, automated security analysis of iOS and Android apps coded in any language
ConfigureCode scanning

ConfigureCode scanning

Publish Node.js Package to GitHub Packages
By GitHub Actions
Publishes a Node.js package to GitHub Packages.
ConfigureJavaScript

ConfigureJavaScript

Publish Node.js Package
By GitHub Actions
Publishes a Node.js package to npm.
ConfigureJavaScript

ConfigureJavaScript

Node.js
By GitHub Actions
Build and test a Node.js project with npm.
ConfigureJavaScript

ConfigureCode scanning

Sysdig Inline Scan
By Sysdig
Performs analysis on locally built container image and posts the results in SARIF report
ConfigureCode scanning

ConfigureCode scanning

Red Hat CodeReady Dependency Analytics
By Red Hat
Scan your project's dependencies with CodeReady Dependency Analytics.
ConfigureCode scanning

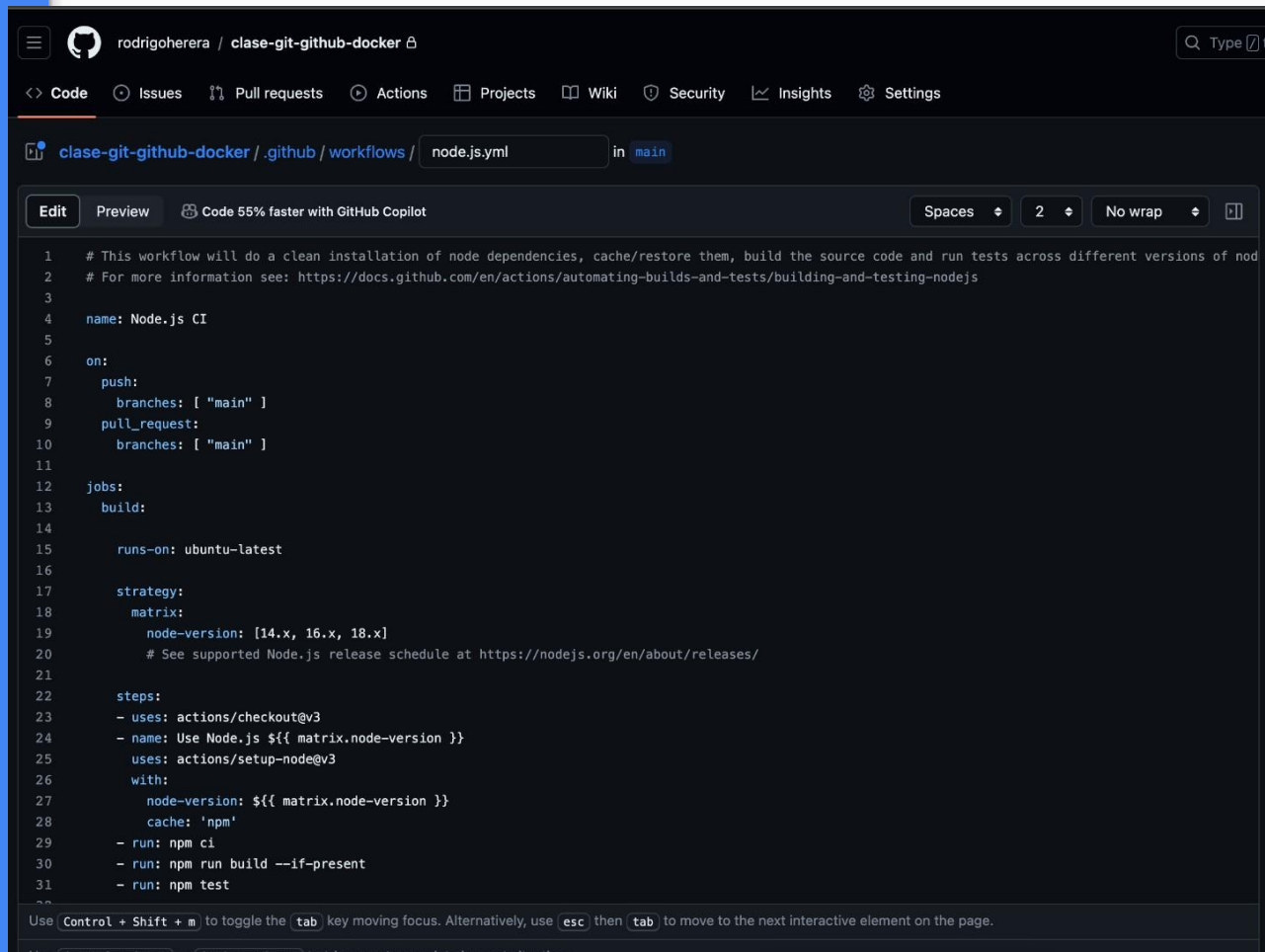
YAML

Nos va a redirigir a una página de edición de un archivo .yml donde está cargada toda la configuración básica para correr un pipeline de Node.js, el mismo incluye:

- Descargar dependencias
- Buildear la aplicación
- Correr los tests

No hace falta modificar nada, solo hacemos commit.

Se crea en nuestra rama principal, una carpeta .github/workflows con un archivo node.js.yml



```
1 # This workflow will do a clean installation of node dependencies, cache/restore them, build the source code and run tests across different versions of nod
2 # For more information see: https://docs.github.com/en/actions/automating-builds-and-tests/building-and-testing-nodejs
3
4 name: Node.js CI
5
6 on:
7   push:
8     branches: [ "main" ]
9   pull_request:
10    branches: [ "main" ]
11
12 jobs:
13   build:
14
15     runs-on: ubuntu-latest
16
17     strategy:
18       matrix:
19         node-version: [14.x, 16.x, 18.x]
20         # See supported Node.js release schedule at https://nodejs.org/en/about/releases/
21
22     steps:
23     - uses: actions/checkout@v3
24     - name: Use Node.js ${{ matrix.node-version }}
25       uses: actions/setup-node@v3
26       with:
27         node-version: ${{ matrix.node-version }}
28         cache: 'npm'
29     - run: npm ci
30     - run: npm run build --if-present
31     - run: npm test
```

Dockerfile multi-stage

Dockerfile multi-stage

Dockerfile multi stage

En el primer stage lo que hacemos es, desde una versión completa de node, creamos un directorio de trabajo, copiamos todos los paquetes package.json, corremos una versión más rápida de npm install, instalando solo los paquetes necesarios para producción y copiamos el file index.js hacia el root de nuestro directorio de trabajo.

En la segunda stage, usamos una imagen básica de node y Debian para que la imagen final del contenedor tenga lo necesario para correr la aplicación y tenga un peso mínimo. Creamos variables de entornos que son necesarias en la aplicación y luego copiamos desde la imagen del stage 1, el directorio de trabajo que creamos, seteamos nuevamente un nuevo directorio de trabajo y corremos el archivo principal.

```
# First stage: Build
FROM node:18.17.1 as build

WORKDIR /app

# Copy only the package.json and package-lock.json first to leverage Docker cache
COPY package*.json ./

# Install dependencies
RUN npm ci --only=production

COPY index.js ./

# Second stage: Run
FROM gcr.io/distroless/nodejs18-debian11

# Group ARGs and ENVs
ARG DB_HOST
DB_USER DB_PASSWORD DB_DATABASE
ENV DB_HOST=$DB_HOST DB_USER=$DB_USER DB_PASSWORD=$DB_PASSWORD DB_DATABASE=$DB_DATABASE

# Copy the app from the build stage
COPY --from=build /app /app

WORKDIR /app

CMD ["index.js"]
```

Docker Hub

Docker Hub

1. Crear una cuenta de docker hub.
2. Crear API KEY

[Account Settings](#) / [Security](#)



developerdocker

user Joined March 7, 2017

General

Security

Default Privacy

Notifications

Convert Account

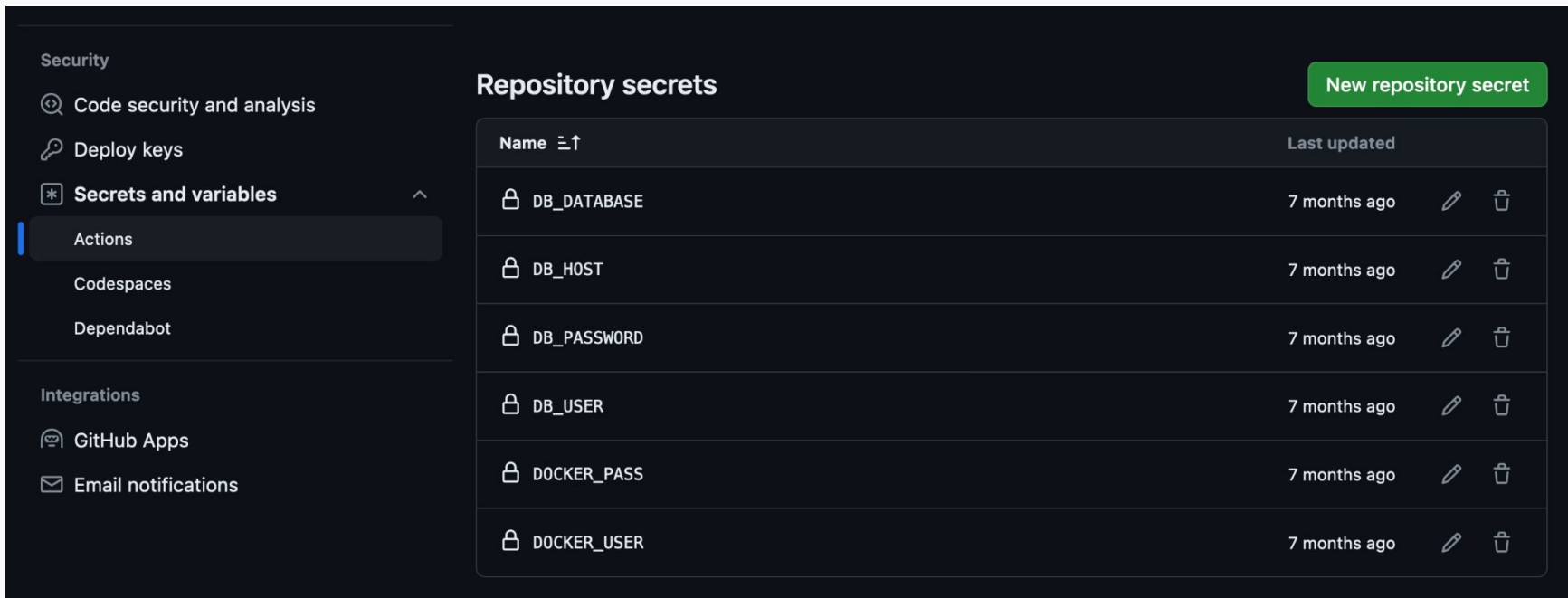
Deactivate Account

Access Tokens

[New Access Token](#)

☐	Description	Source	Scope	Last Used		Created		
☐	github action node-servi...	MANUAL	Read, Write, Delete	Nov 15, 2023	19:41:44	Sep 24, 2023	19:50:09	⋮
☐	recipe-app-api	MANUAL	Read, Write, Delete	Nov 26, 2022	10:51:26	May 20, 2022	13:45:52	⋮

Docker Hub - secretos github



The screenshot shows the GitHub interface for managing repository secrets. On the left is a sidebar with navigation options under 'Security' and 'Integrations'. The 'Secrets and variables' section is expanded, showing 'Actions' as the selected option. The main area is titled 'Repository secrets' and features a table of existing secrets. A green button labeled 'New repository secret' is in the top right corner of the main area. The table lists six secrets: DB_DATABASE, DB_HOST, DB_PASSWORD, DB_USER, DOCKER_PASS, and DOCKER_USER. Each entry shows the secret name, its last update time (all are '7 months ago'), and icons for editing or deleting the secret.

Security

- Code security and analysis
- Deploy keys
- Secrets and variables**
- Actions
- Codespaces
- Dependabot

Integrations

- GitHub Apps
- Email notifications

Repository secrets

New repository secret

Name ↕	Last updated
DB_DATABASE	7 months ago
DB_HOST	7 months ago
DB_PASSWORD	7 months ago
DB_USER	7 months ago
DOCKER_PASS	7 months ago
DOCKER_USER	7 months ago

Docker Hub

En esta etapa interactuamos con el Docker Hub

Nos logueamos con user y password (para el password usamos el Access Token)

Obtenemos metadata de la imagen del hub.

Usando el file de docker multi stage, agregamos el tag de la nueva versión y le pasamos argumentos extras

```
28   push_to_docker_hub:
29     name: Push Docker image to Docker Hub
30     runs-on: ubuntu-latest
31     needs: build_and_test
32     if: github.ref == 'refs/heads/main' && github.event_name == 'push'
33
34     steps:
35       - name: Check out repo
36         uses: actions/checkout@v3
37       - name: Log in to Docker Hub
38         uses: docker/login-action@v2
39         with:
40           username: ${ secrets.DOCKER_USER }
41           password: ${ secrets.DOCKER_PASS }
42       - name: Extract metadata (tags, labels) for Docker
43         id: meta
44         uses: docker/metadata-action@v4
45         with:
46           images: ${ github.repository }
47       - name: Build and push Docker image
48         uses: docker/build-push-action@v4
49         with:
50           context: .
51           file: Dockerfile
52           push: true
53           tags: ${ secrets.DOCKER_USER }/user-service:${ github.sha }
54           labels: ${ steps.meta.outputs.labels }
55           build-args: |
56             DB_HOST=${ secrets.DB_HOST }
57             DB_USER=${ secrets.DB_USER }
58             DB_PASSWORD=${ secrets.DB_PASSWORD }
59             DB_DATABASE=${ secrets.DB_DATABASE }
```

Deploy a Digital Ocean

Deploy a Digital Ocean

En esta etapa, vamos a usar la imagen ya creada, como argumento para interactuar por SSH directamente con nuestro servidor de DigitalOcean, donde vamos a ejecutar un script con el parámetro extra que sería nuestra nueva versión de la aplicación.

<https://github.com/rodrigoherera/clase-git-github-docker>

<https://github.com/lucasbonanni/clase-git-github-docker>

```
deploy_to_digitalocean:
  name: Deploy to digital ocean
  runs-on: ubuntu-latest
  needs: push_to_docker_hub_registry

  steps:
  - name: executing remote ssh commands using password
    uses: appleboy/ssh-action@v1.0.0
    with:
      host: ${ secrets.HOST }
      username: ${ secrets.USERNAME }
      key: ${ secrets.KEY }
      port: ${ secrets.PORT }
      passphrase: ${ secrets.PASSPHRASE }
      script:
        "/bin/bash /root/docker/deploy.sh ${ github.sha }"
```

Render (render.com)

Crear cuenta y crear web services

Sign up for Render



GitHub



GitLab



Google

OR

Email

your@email.com

Password

correct horse battery staple



Email me about Render product updates

COMPLETE SIGN UP

By signing up you agree to our [terms of service](#).

Already have an account?

[Sign in](#)

The screenshot shows the Render dashboard at dashboard.render.com. The user is logged in as Yogesh Chavan. The dashboard features an 'Overview' section titled 'Get up and running in minutes' with a grid of service cards. The 'Web Services' card is highlighted with a green border. Other cards include Static Sites, Private Services, Background Workers, Cron Jobs, PostgreSQL, Redis, and Blueprints. Each card provides a brief description and a 'New' button.

Service	Description	Action
Static Sites	Static sites are automatically served over a global CDN. Add a custom domain and get free, fully-managed SSL.	New Static Site
Web Services	Web services include zero-downtime deploys, persistent storage and PR previews. Scale up and down with ease.	New Web Service
Private Services	Private services are only accessible within your Render network and can speak any protocol.	New Private Service
Background Workers	Background workers are suitable for long running processes like consumers for queues and streaming.	New Worker
Cron Jobs	With cron jobs you can schedule any command or script to run on a regular interval.	New Cron Job
PostgreSQL	Fully-managed hosted PostgreSQL with internal and external connectivity, and automated daily backups.	New PostgreSQL
Redis	A cloud based in-memory key value datastore. Render offers fully managed hosted Redis instances.	New Redis
Blueprints	A Blueprint specifies your Infrastructure as Code in a single file. Use it to set up all your services at once.	New Blueprint

Services

From

\$0 USD

per month

Show Pricing

- ✓ Web services with HTTP/2 and full TLS
- ✓ Private services
- ✓ Background workers

- ✓ Node, Python, Go, Rust, Ruby, and Elixir
- ✓ Custom Docker containers
- ✓ SSDs for \$0.25/GB per month

Conectar la cuenta y free tear



Install Render

Install on your personal account Yogesh Chavan

☐ All repositories

This applies to all current and future repositories.

☒ Only select repositories

Select at least one repository.

Select repositories

with these permissions:

- ✓ Read access to Dependabot alerts, administration, code, and metadata
- ✓ Read and write access to actions, checks, commit statuses, deployments, environments, issues, pull requests, repository hooks, and workflows

User permissions

Render can also request users' permission to the following resources. These permissions will be requested and authorized on an individual-user basis.

Install on your personal account Yogesh Chavan

☐ All repositories

This applies to all current and future repositories.

☒ Only select repositories

Select at least one repository.

Select repositories

Selected 1 repository.

myogeshchavan97/github-repos-nodejs-api

with these permissions:

- ✓ Read access to Dependabot alerts, administration, code, and metadata
- ✓ Read and write access to actions, checks, commit statuses, deployments, environments, issues, pull requests, repository hooks, and workflows

User permissions

Render can also request users' permission to the following resources. These permissions will be requested and authorized on an individual-user basis.

- ✓ Read access to emails

Install Cancel

Next: you'll be directed to the GitHub App's site to complete setup.

render

Dashboard

Blueprints

Env Groups

Free

RAM 512 MB
CPU shared

\$0/month

Starter

RAM 512 MB
CPU 0.5

\$7/month

Pro

RAM 4 GB
CPU 2

\$85/month

Pro Plus

RAM 8 GB
CPU 4

\$175/month

Unlike paid plans, free services scale down when inactive. They also

Need a custom plan? We support u

Advanced

Create Web Service

Copiar la URL del Hook

[Dashboard](#)[Blueprints](#)[Env Groups](#)[Docs](#)[Community](#)[Help](#)[New +](#)

Lucas

[Events](#)[Logs](#)[Disks](#)[Environment](#)[Shell](#)[Previews](#)[Jobs](#)[Metrics](#)[Scaling](#)[Settings](#)

Docker Command

Add an optional command to override the Docker `CMD` for this service. This will also override the `ENTRYPOINT` if defined in your Dockerfile. Examples: `./start.sh --type=worker` or `./bin/bash -c cd /some/dir && ./start.sh`

[Edit](#)

Pre-Deploy Command Optional

This command runs before starting your service. It is typically used for tasks like running a database migration or uploading assets to a CDN.

[Edit](#)

Deploy Hook

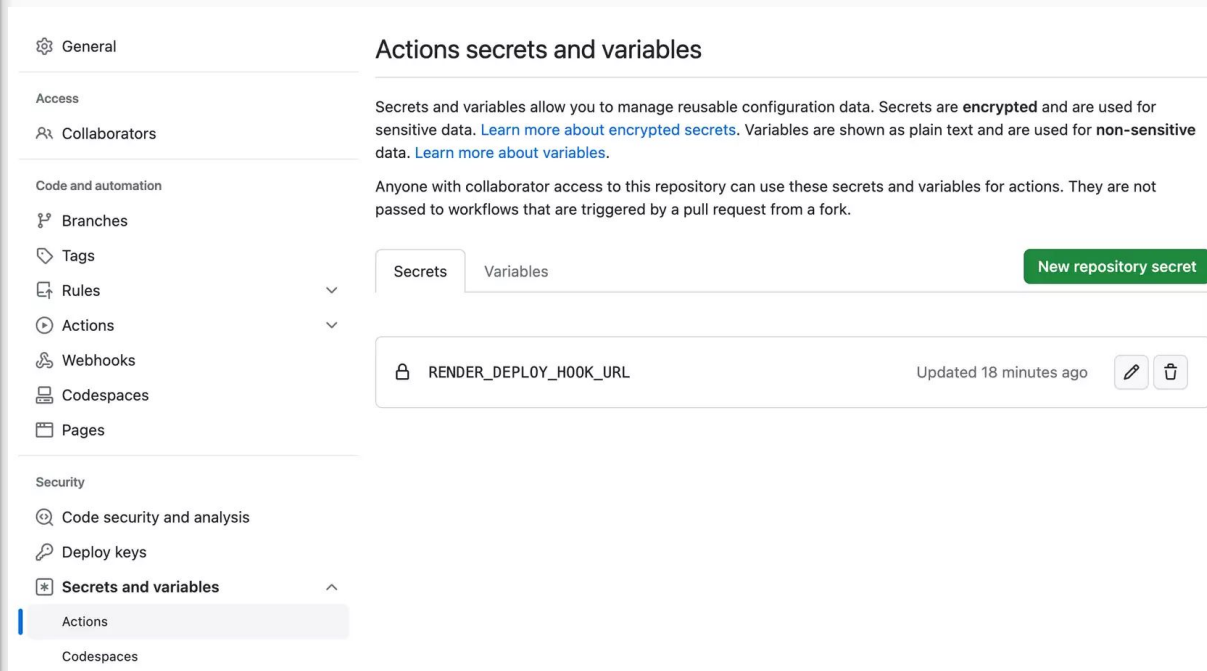
Your private URL to trigger a deploy for this server. Remember to keep this a secret.

`https://api.render.com/deploy/srv-c`[Regenerate Hook](#)

1. Crear un repositorio secret

Las URL de los hooks de despliegue son valores secretos, así que tenemos que asegurarnos de almacenar la nuestra como secreta en nuestro repositorio de GitHub:

1. Ve a la página de Configuración de tu repositorio de GitHub.
2. Haz clic en Secretos y variables > Acciones.
3. Haz clic en Nuevo secreto de repositorio. Crea un secreto con el nombre `RENDER_DEPLOY_HOOK_URL` y proporciona la URL de tu hook de despliegue como valor.



The screenshot shows the GitHub interface for managing repository secrets and variables. On the left is a sidebar with navigation options: General, Access, Collaborators, Code and automation (with sub-items: Branches, Tags, Rules, Actions, Webhooks, Codespaces, Pages), Security (with sub-items: Code security and analysis, Deploy keys, Secrets and variables), Actions, and Codespaces. The 'Secrets and variables' section is expanded, showing 'Secrets' and 'Variables' tabs. The 'Secrets' tab is active, displaying a list of secrets. One secret is visible: 'RENDER_DEPLOY_HOOK_URL', which is locked (indicated by a padlock icon) and was updated 18 minutes ago. To the right of the secret name are edit and delete icons. A green button labeled 'New repository secret' is located in the top right corner of the secrets list area. The main content area on the right contains explanatory text about secrets and variables, stating that secrets are encrypted and used for sensitive data, while variables are shown as plain text and used for non-sensitive data. It also notes that secrets and variables are available to anyone with collaborator access to the repository.

General

Access

Collaborators

Code and automation

- Branches
- Tags
- Rules
- Actions
- Webhooks
- Codespaces
- Pages

Security

- Code security and analysis
- Deploy keys
- Secrets and variables

Actions

Codespaces

Actions secrets and variables

Secrets and variables allow you to manage reusable configuration data. Secrets are **encrypted** and are used for sensitive data. [Learn more about encrypted secrets](#). Variables are shown as plain text and are used for **non-sensitive** data. [Learn more about variables](#).

Anyone with collaborator access to this repository can use these secrets and variables for actions. They are not passed to workflows that are triggered by a pull request from a fork.

Secrets Variables

New repository secret

RENDER_DEPLOY_HOOK_URL Updated 18 minutes ago

Crear el workflow

.github/workflows/ci.yml

YAML

on:

pull_request:

push:

branches: [main]

jobs:

ci:

runs-on: ubuntu-latest

steps:

- uses: actions/checkout@v3

- name: Test

run: |

npm install

npm run test

- name: Deploy

Only run this step if the branch is main

if: github.ref == 'refs/heads/main'

env:

deploy_url: \${ secrets.RENDER_DEPLOY_HOOK_URL }

run: |

curl "\$deploy_url"

Next Security